

BRIEF REPORT

Genomic architecture and breeding trade-offs of coconut drought tolerance

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[Correction added on 3 September 2026, after first online publication: The eleventh author's name has been corrected to 'Hala N'klo'.]

Societal Impact Statement

The coconut palm is an extremely popular and economically important crop, with around 65 million tonnes produced in 2023. Although Africa's coconut sector is growing rapidly, it—and the livelihoods that depend on it—is threatened by climate change, particularly increased drought. Although developing stress-resilient crops remains challenging, here we aimed to first understand the variation in drought response and then to dissect the genomic basis of this complex trait, leveraging an experimental system in Côte d'Ivoire. Despite identifying a trade-off between drought tolerance and yield, we propose that genomic selection, after validation, could help expand production onto marginal land.

KEYWORDS

coconut palm, crop production, drought tolerance, palm genomes, sustainable agriculture

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1 | INTRODUCTION

Identifying genes underlying stress tolerance is increasingly urgent due to climate change, soil degradation and the need to increase yields on marginal lands. While exceptions such as salt-tolerant quinoa exist (Zou et al., 2017), developing stress-resilient crops remains difficult. Domestication has often narrowed genetic diversity, as seen in maize (Hufford et al., 2012), and stress-related traits, such as salt tolerance in rice (Kumar et al., 2015) and drought tolerance in maize (Wallace et al., 2014), are typically polygenic. Despite their ecological and economic importance, palms have been largely neglected in genomics. Coconut palms (*Cocos nucifera* L.) are economically important, with over 65 million tonnes of coconuts produced in 2023 across nearly 90 countries (Food and Agricultural Organization of the United Nations, 2026). Indonesia, the Philippines and India each produce ~3 million tonnes annually, followed by Brazil with 2.8 million (Rodrigues et al., 2018). While Asia leads in production, Africa's coconut sector is growing rapidly. Ghana, for example, aims to plant one million hectares, creating 200,000 jobs and generating USD 12 billion (Oil Palm Research Institute, 2026). Côte d'Ivoire, the continent's top exporter, already cultivates over 40,000 ha, producing more than 120,000 t in 2023 (Food and Agricultural Organization of the United Nations, 2026). Climate change, however, poses a major threat to future palm agriculture. Across other palms from the wet tropics, projections suggest over 70% of Africa's palm-growing areas may become climatically unsuitable by 2080 (Blach-Overgaard et al., 2015), with 42% of the most climate-affected areas also occurring in densely populated regions that rely on these crops. In the face of mounting climate stress, how can we secure a resilient future for this globally important crop?

Focusing on drought as a major environmental stressor, here, we combine phenotypic and genomic analyses to understand drought tolerance in coconut palm and inform future breeding efforts in this crop. Specifically, we analysed yield variation in the progeny of hybrid crosses between major varieties (parental lines) in relation to climatic conditions. These crosses are maintained at the Marc Delorme Research Station, Centre National de Recherche Agronomique in Côte d'Ivoire, host of the world's largest coconut collection (International Coconut Genebank for Africa and the Indian Ocean) with over 22,000 trees from 57 ecotypes spanning from across the world. Established over 50 years ago, this breeding program has produced varieties with much higher yields than more traditional varieties, such as the West African Tall (WAT), but often with increased drought sensitivity (Konan et al., 2006). Understanding this trade-off, as well as the genetic basis of drought tolerance itself, is therefore essential for informing future breeding efforts.

2 | METHODS

The full description of the methodology is available in Methods S1; below is a summary, expanded on where relevant in Section 3.

Before modelling tree-level drought responses, we screened four drought metrics, each across three timescales, using data from 4094

trees representing crosses between coconut varieties from a broad geographic range spanning West Africa to the Pacific and Central America (e.g., Table S1). We used the best performing drought metric to quantify drought sensitivity using a two-stage reaction norm framework: first adjusting each tree's annual yield for year and age effects unrelated to drought (stage 1), then regressing these adjusted yields against the centred drought metric separately for each tree with sufficient data, including observations under dry conditions, to estimate its individual quadratic response to moisture availability (stage 2). We then summarised each tree's response using the linear approximation of the quadratic slope over typical drought conditions, ultimately deriving a single drought sensitivity phenotype (hereafter, the 'drought slope estimate') for each tree.

To explore the genomic basis of this variation, we sequenced 376 coconut trees spanning the drought tolerance-sensitivity spectrum, including replicates of each cross (Tables S3 and S4). We then estimated the SNP-based heritability (h^2_{SNP}) of the drought slope estimate and performed a genome-wide association study (GWAS) to identify associated loci, alongside genetic diversity and population structure analyses. Finally, we calculated breeding values and polygenic risk scores for individual trees and parental lines to inform breeding recommendations.

3 | RESULTS AND DISCUSSION

3.1 | Modelling responses to drought at the individual tree-level

To select the best way to model tree-level variation in drought response, we first used a mixed-modelling approach to screen four different drought metrics across three timescales with and without the addition of a quadratic term. The best-performing model was that using the 18-month reconnaissance drought index (RDI-18)—the ratio of total rainfall to potential evapotranspiration accumulated over a rolling 18-month window, with lower values indicating drier conditions—where the inclusion of a quadratic term substantially improved model fit (Table S2, see Methods S1 for further details of each drought metric).

We quantified drought sensitivity at the individual-tree level using a reaction norm framework (Escobar-Sandoval et al., 2021; Mangin et al., 2017), modelling yield responses against RDI-18 while accounting for year and tree age. Consistent with the substantial inter-year variation in yield we observed in the raw data (mean annual nut count per tree $\sim 13.6 \pm 7.4$ in the main dataset for the 28-year study period), we found similar variation in reaction norm intercepts (i.e., baseline yields at RDI-18 values of zero) (13.5 ± 1.6). We also observed divergent drought responses (Figure 1a). Unlike controlled watering experiments (Ferreira & Gonçalves, 2007; Gheysari et al., 2021; Yazar et al., 2009), our model accounts for environmental variation and showed that around half of the coconut palms had a negative quadratic response and thus were sensitive to both drought and waterlogging, in line with findings from maize (Kipkorir

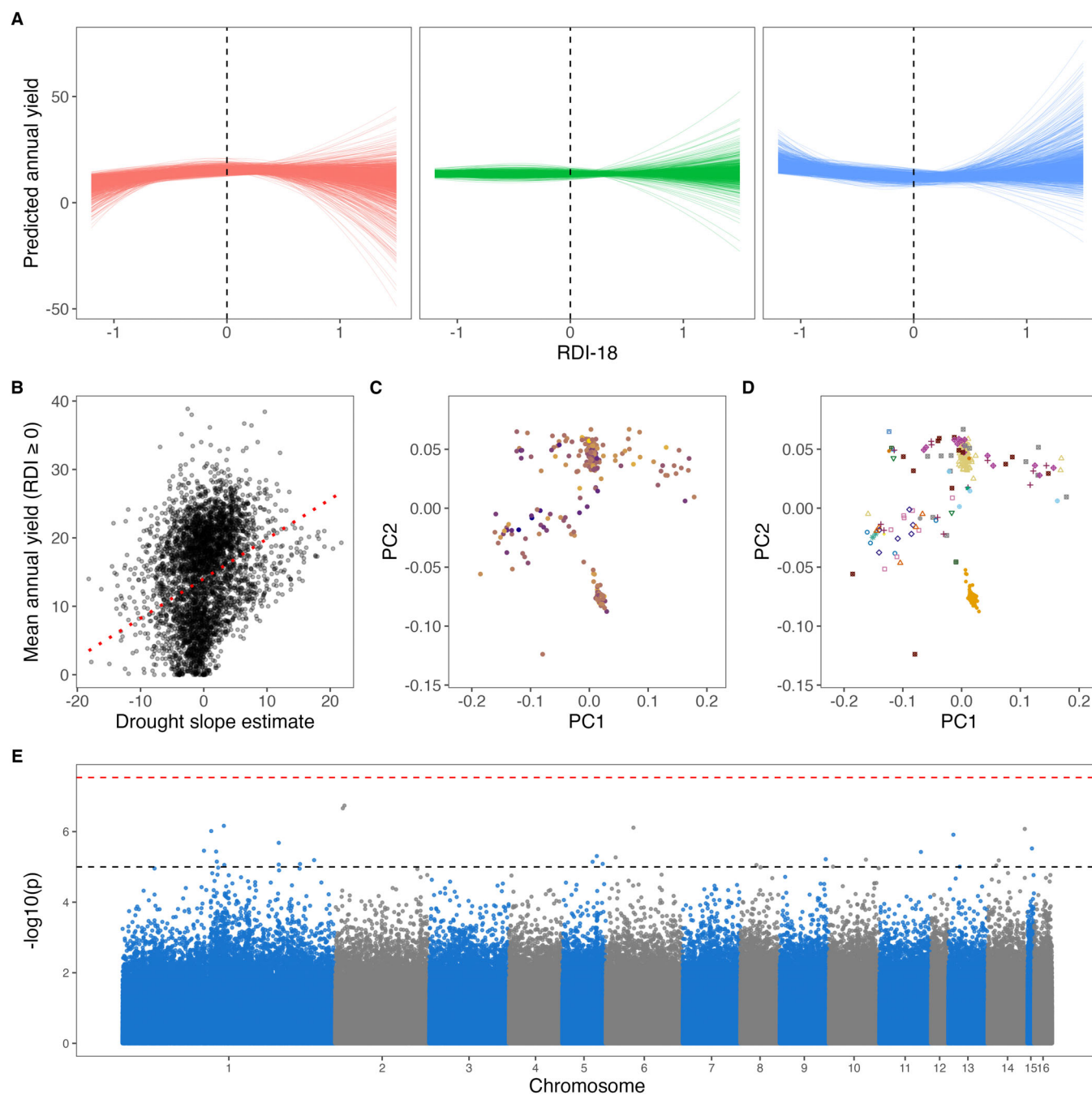


FIGURE 1 Legend on next page.

et al., 2002). Notably, the remaining palms showed positive quadratic trends, indicating higher yields under both dry and wet conditions (Figure 1a). To allow us to compare individual trees, we summarised drought responses using the linear approximation of the quadratic slope over typical drought conditions ($-1 \leq \text{RDI-18} \leq 0$), the drought slope estimate. A larger drought slope estimate means that yield increases as moisture levels increase (i.e., greater drought sensitivity), whereas smaller or negative drought slope estimate indicates more stable yield across drought levels or even increasing yield with decreasing moisture levels (i.e., greater drought tolerance). Correlations between drought slope estimates calculated from across the different drought metrics were moderate (mean Pearson correlation coefficient = 0.57) (Table S5), highlighting that our estimations of drought tolerance are sensitive to the specific drought metric used and thus our above metric selection step is important.

Alongside variation in yield responses to drought, we also identified a positive correlation between mean yield and the drought slope estimate (Figures 1b and S1). As these drought slope estimates are inversely related to drought tolerance—where more positive values indicate lower tolerance—this correlation suggests a modest trade-off between yield potential and tolerance, consistent with patterns reported in other crops (Gao et al., 2022; Tardieu et al., 2018). Given this phenotypic variation, below, we aim to dissect the contributions of genomic versus environmental factors and thus support breeding for improved drought tolerance.

3.2 | Uncovering the genomic basis of drought tolerance

To explore the genomic basis of drought tolerance and breeding potential of different hybrid crosses, we used 1,702,034 SNPs obtained from genome sequencing (see Methods S1) to estimate the h^2_{SNP} to assess the genomic basis of drought tolerance and performed

a genome-wide association study (GWAS) to determine if any SNPs were significantly associated with this trait. We first, however, calculated genetic diversity metrics (Table S6), including various numbers of trees depending on support for the drought slope estimate. Initial assessment of population structure revealed a mean inbreeding coefficient of 0.13, indicating heterozygote deficit, a surprising observation for hybrid crosses but consistent with the population stratification supported by strong structure seen in PCA (Figures 1c,d, S2 and S3 and Table S7). This complex population structure and high genetic diversity highlight the challenges and opportunities for association mapping and crop improvement in this species.

Given the population structure by cross type and drought response (Figures 1c,d, S2 and S3 and Table S7), we included principal components as covariates in the following models. For the 321 trees we could accurately calculate drought slope estimates for, we calculated genetic and residual variance components of 6.74 (SE 3.94) and 10.02 (SE 3.69), respectively. The h^2_{SNP} was 0.40 (SE 0.23), showing additive genetic effects captured by common SNPs explain a large proportion of the phenotypic variance in drought tolerance (Table S8). This large h^2_{SNP} is consistent with or exceeds broad-sense heritability estimates reported in other crops (Table S9) suggesting a substantial genetic component to this trait. Furthermore, h^2_{SNP} was positively correlated with chromosome size (Figure S4A), suggesting that larger chromosomes contribute proportionally more to heritability and thus a polygenic architecture. This relationship was, however, driven by chromosome 1, indicating this chromosome contributes disproportionately to total heritability (Figure S4B).

After confirming a genetic component, we performed the GWAS. Although the genome inflation factor indicated a well-calibrated test ($\lambda = 0.99$) (Figure S5), we did not identify any genome-wide significant SNPs when using a strict Bonferroni-corrected threshold of 2.93×10^{-8} . We did, however, identify 28 associated SNPs when using a less stringent threshold of 1×10^{-5} (Figure 1e and Table S10). Although these SNPs had relatively large effect sizes (absolute effect

FIGURE 1 Summary of the phenotypic variation in drought response among individual coconut palms (a), the associated trade-off between drought tolerance and yield (b), and the genomic architecture underlying this variation, identified through population structure analysis (c, d) and a genome-wide association study (e). (a) Per-tree quadratic reaction norms of adjusted annual nut yield (number of nuts) against the reconnaissance drought index measured over an 18-month period (RDI-18) for a subset of individual trees with the most positive (left, red, $n = 300$), flattest (middle, green, $n = 300$), and most negative (right, blue, $n = 300$) mean drought slope estimates, drawn from a dataset of 4094 trees. After adjusting yield for tree age (fixed effect) and year and tree identity (random effects) using a linear mixed model, we re-centred residuals on the grand mean and fitted a regression—or reaction norm—of adjusted yield on RDI-18 independently for each tree. We then calculated the drought slope estimates by extracting the linear approximation of the quadratic slope between RDI-18 values of -1 and 0 , meaning a larger drought slope estimate indicates greater drought sensitivity while a smaller or negative drought slope estimate indicates greater drought tolerance. A broadly flat drought slope estimate indicates insensitivity of a given palm's response to drought. We plotted the curves of predictions over the range RDI-18 range each tree experienced. (b) Scatter plot showing the observed mean annual nut yield in wetter-than-average years ($\text{RDI-18} \geq 0$) against the model-derived drought slope estimate. The positive relationship (Pearson correlation coefficient = 0.27, $p < 0.001$) indicates that more drought sensitive trees (i.e., trees with a more positive drought slope estimate) tend to have higher yields, while more drought tolerant trees (i.e., more negative drought slope estimate) tend to have lower yields—suggesting a moderate trade-off between yield and drought tolerance. (c, d) Principal components 1 (PC1) and 2 (PC2) summarising genome-wide SNP variation and coloured according to (c) drought slope estimate (gradient: blue = more positive drought slope estimate, yellow = more negative drought slope estimate; full figure in Figure S2) and (d) cross type (different colours/shapes indicate different crosses; full figure in Figure S3). (e) Manhattan plot depicting results of the genome-wide association study (GWAS) ($n = 1,702,034$ SNPs). SNPs with p -values above the upper, red threshold are defined as significant ($n = 0$ significant SNPs) and SNPs with p -values above the lower, black threshold are defined as suggestive ($n = 28$ suggestive SNPs).

sizes = 2.10 to 6.42; variance explained = 9.48%–19.37%) (Table S10), these per-SNP estimates are likely markedly inflated due to an upward bias in effect size estimation when weakly associated SNPs are identified based on their test statistic in a relatively small sample sizes ($n = 321$) (Göring et al., 2001; Palmer & Pe'er, 2017). The absence of any genome-wide significant associations despite a significant h^2_{SNP} , alongside the dispersion of suggestive SNPs across chromosomes, is consistent with a polygenic architecture, where many small effect loci collectively contribute to drought tolerance. This pattern mirrors findings in sesame (Berhe et al., 2021) and *Eucalyptus* (Müller et al., 2017), where single-locus GWAS identified few variants but genome-wide approaches captured a high heritability. Given single-locus methods can miss complex genetic effects due to multiple testing, pleiotropy and epistasis (Bush & Moore, 2012; Gawenda et al., 2015), we recommend future work use larger sample sizes and multi-locus models like *gBLUP* (Clark & van der Werf, 2013) to better capture polygenic signals. We propose that a genome-wide SNP panel could be used for genomic selection of drought tolerant coconut varieties (e.g., Stocks et al., 2019); however, this requires further validation in independent populations of trees with characterised drought tolerance.

Functional annotation of the suggestive SNPs (Table S11) revealed that 26 of these suggestive SNPs are intergenic, while another on chromosome two is upstream of an uncharacterised protein-coding gene and may therefore have a regulatory role. The remaining SNP, located on the fifth chromosome, is within a gene encoding a putative E3 ubiquitin-protein ligase, a regulator of protein turnover and posttranslational modification. Related proteins in *Arabidopsis* mediate stress-induced metabolic adjustments and E3 ubiquitin ligases regulate protein degradation during stress signalling (Stone, 2014). This does, however, require functional validation and another GWAS with a greater sample size is required to confirm this association is genuine.

3.3 | Practical implications for farmers

We calculated breeding values for individual trees to guide breeders. Hybrid trees with parents from certain lines, such as LMT7 and VTT, tended to show higher breeding values and polygenic risk scores for drought tolerance (Figures S6 and S7). The highest scoring individuals were, however, spread across crosses (Figures S6 and S7). Genomic data from eight parental varieties also revealed that many individual parental lines had between 25% and 50% of alleles positively associated with drought tolerance and two varieties (VTT and CRD) carried alleles found in no other sequenced line (Table S12). This dispersion of high breeding values and more favourable alleles, alongside the observed heterozygosity and our incomplete data for some parental lines known for drought tolerance (e.g., WAT) (Konan et al., 2006), suggests that individual tree-level genomic selection within crosses could compliment cross-level selection.

We highlight the complexity and variability of drought response in coconut palms, showing curvilinear relationships between moisture

and yield. Although other yield variables, such as nut size and quality, must be considered in any breeding programmes, we identify a potential trade-off between drought tolerance and the number of nuts produced. We recognise that few growers would choose drought-tolerant varieties for fertile areas; however we propose that genomic selection and breeding of drought-tolerant varieties could expand coconut cultivation into marginal/arid regions, delivering sustainable agricultural and economic gains. Crucially, this requires validation in independent populations before implementation. More broadly, our study underscores the importance of long-term yield records in long-lived crops like coconut. Sustained data collection, combined with large-scale genomic analyses, will be crucial for developing climate-resilient tree crops that underpin sustainable food security in the tropics and beyond.

AUTHOR CONTRIBUTIONS

Vincent Savolainen, Theodore S. Brook, Bi Tra Serges Doubi, Jean Louis Konan, Matteo Fumagalli and Tom Versluys designed the research; Bi Tra Serges Doubi, Jean Louis Konan, Konan Engueran Djaha, Jahcub Trew, Benjamin J. Roberts and Catherine M. (Tilly) Collins collected samples and data; Theodore S. Brook conducted most of the analyses with contributions by Sam Acors, Jahcub Trew, Kyle Macleod, Louis Bell-Roberts and Katarina Piponi; Theodore S. Brook wrote the initial draft manuscript with Vincent Savolainen; all authors commented on the text; Vincent Savolainen supervised the research.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ETHICS STATEMENT

There are no ethical declarations.

DATA AVAILABILITY STATEMENT

All climate data, yield data, and code are available at <https://github.com/SavolainenLab/Brook-et-al-2026-Cocos>. All genomic data are available under the project PRJNA1435638, available at <https://www.ncbi.nlm.nih.gov/bioproject/1435638>.

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